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Area-based measures of socioeconomic status in studies assessing health outcomes among people living with HIV in Canada and the USA: a scoping review

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ABSTRACT

Objective Area-based measures of socioeconomic status (SES) are increasingly used to study health disparities among people living with HIV (PLWH), with wide variation in how they are defined and applied across studies and settings. This study synthesises the types of area-based measures of SES used in Canada and the USA, the domains captured and their associations with health outcomes among PLWH.

Methods A scoping review of studies published in English between 2012 and 2025 was conducted using PubMed and Web of Science. The search combined 'HIV' with terms related to area-based SES measures. Eligible studies included PLWH, were based in Canada or the USA, used area-based SES measures and assessed health outcomes.

Results We screened 3470 studies: 56 met inclusion criteria. Most were US-based (n=53) and focused solely on PLWH (n=46). Area-based SES was measured using composite (n=34), single (n=16) or both types of indicators (n=6), all drawn from census data. The most common SES domain was income/poverty (n=56), the most common geographic unit was census tract (n=19) and the most common health outcome assessed was viral load/suppression (n=29). Most studies linked lower area-based SES with poorer health outcomes among PLWH (n=46).

Conclusions Our findings highlight the utility of area-based SES as an individual-level SES proxy and tool for capturing broader social determinants of health when assessing a range of health outcomes across studies including PLWH. This review contributes to strengthening methodological approaches and supports future work focused on addressing social determinants and advancing health equity for PLWH.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ A key strength of this scoping review is its compliance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines, which provide a uniform systematic structure for reporting search, screening and data extraction processes and results.
⇒ A comprehensive search strategy was applied across two major databases (PubMed and Web of Science), to capture a broad scope of relevant literature related to HIV and area-level measures of socioeconomic status (SES).
⇒ Restricting inclusion to studies of people living with HIV and their health outcomes allowed for a nuanced and in-depth analysis of area-based measures of SES used in this specific context.
⇒ The restricted focus on studies from Canada and the USA may limit generalisability to other global settings with different healthcare systems and/or socioeconomic structures.

INTRODUCTION

The global rate of new HIV infections has been continuously decreasing worldwide due to the success of antiretroviral therapy (ART).[1] As antiretroviral regimens improve and become more widely accessible, people living with HIV (PLWH) are living longer.[2 3] However, as more PLWH live into advanced years, there is a rising burden of non-AIDS related health conditions, which are becoming increasingly important to address in HIV care.[4] Concurrently, disparities in health outcomes among PLWH are widening, partly driven by the social determinants of health.[5–7]

Socially marginalised and economically disadvantaged individuals face a higher risk of acquiring HIV, and among PLWH, social determinants of health play a critical role in shaping both short and longer-term health outcomes.[8] The WHO defines the social determinants of health as the conditions responsible for the 'unfair and avoidable differences in health status seen both within and between countries.'[9] Socioeconomic status (SES) represents a key component of the social determinants of health, typically encompassing factors such as income, education and employment. However, SES is also closely linked to other intersecting dimensions of social and physical environments, including housing stability, social support networks, access to health services and other aspects of the physical living environment.[9] Capturing SES in data with validity and accuracy remains a public health priority, but it is challenging due to its multidimensional nature and overlap with broader social determinants of health.[9]

Rationale for using area-based SES measures

There is a growing use of routinely collected data, such as administrative health records and electronic medical records, in HIV research.[10 11] These data sources offer a practical way to study health outcomes at the population level, particularly in large-scale studies.[12] However, such datasets often lack comprehensive and accurate SES information.[13] Researchers conducting primary data collection through cohort studies or surveys can include more detailed SES measures, though this information may be limited by participation rates, missing data or self-report bias.

To address these limitations, area-based SES measures provide an efficient and scalable tool to analyse and quantify socioeconomic differences. These measures are generally less prone to self-report bias and can be linked to health data through geographic identifiers such as ZIP (USA) or postal codes (Canada) or other boundaries associated with residential addresses. When collected consistently, they can also provide longitudinal information, including changes in SES over time.[12] Large cohort studies, such as the Mortality Disparities in American Communities study which consists of over four million individuals, found area-based measures were a fair proxy of individual SES.[14] Additional studies have reported a link between area-based measures and self-reported SES across various geographic levels, including ZIP code, neighbourhood block and census tract.[15] However, it is important to note that in some cases area-based measures can mask heterogeneity of SES and depending on the area used, underrepresent or overrepresent true individual-level SES.[14 16]

Area-based SES measures can also capture broader contextual influences on health, including social, economic and physical conditions of communities within defined areas, which may influence health outcomes independent of individual SES.[15] Area-based SES measures have been linked to perceived neighbourhood quality and characteristics, highlighting their utility in representing an area in terms of how an individual would perceive their surroundings.[17] Interestingly, the relationship between SES and health outcomes can vary depending on how SES is measured and the interaction between individual-level and area-level SES.[18] For example, a large study in China found that low SES individuals living in high-SES counties had the highest risk of schizophrenia.[19] On the other hand, a US study found that among individuals with lower SES, living in areas with higher neighbourhood SES was associated with better cardiovascular health.[20]

Area-based SES measures are typically derived from national census surveys in countries like Canada and the USA. These can be operationalised into simple measures, such as the proportion of individuals below the poverty line in an area, or more complex composite measures such as the Area Deprivation Index (ADI) in the USA and the Material and Social Deprivation Index (MSDI) in Canada.[21 22] Both composite area-based measures of SES combine indicators to quantify disadvantage at an area-level.[21 22]

Increasingly, area-based SES measures are being used to study disparities in health outcomes, including among PLWH. A previous systematic review of studies in high-income countries found significant associations between SES and HIV-related health outcomes.[23] However, this review was limited to HIV-related outcomes such as CD4 count, viral load and ART adherence; it did not examine other comorbidities and health outcomes or focus specifically on area-based SES measures. While many of the 48 studies reviewed identified associations between SES disadvantage and poorer virological and immunological responses, as well as higher rates of ART nonadherence, important gaps remain in understanding how area-based SES measures specifically are applied across a wider range of health outcomes among PLWH.

Objective

Although area-based measures of SES are widely used to examine social determinants of health among PLWH, to our knowledge, no study has comprehensively characterised the types, domains and methodological applications of these measures in relation to health outcomes among PLWH. To address this gap, this scoping review synthesises literature from Canada and the USA examining the association between area-based measures of SES and health outcomes among PLWH. Specifically, this review aims to identify the types of area-based measures used, the underlying socioeconomic domains they represent and whether they are associated with health outcomes in a population. Findings from this review will contribute to the evidence base informing best practices in the use of area-based measures of SES as tools to investigate and address socioeconomic disparities in HIV care and research.

METHODS

Our protocol was informed by the Joanna Briggs Institute Manual for Evidence Synthesis and revised with input from the research team.[24] Findings from this scoping review are reported in agreement with the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR).[25] The protocol has not been registered.

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To be included in this review, studies had to examine the relationship between SES and health outcomes in PLWH using area-based measures. We included peer-reviewed articles published in English between 1 January 2012 and 31 December 2025, conducted in Canada and/or the USA. The publication date range was selected to focus on recent, relevant literature. We limited our search to English-language studies to align with the language proficiency of all reviewers.

We restricted our review to studies conducted in Canada and the USA based on contextual similarities between the two countries. Both face a similar HIV/AIDS epidemic marked by racial and ethnic disparities and emphasise public health strategies focused on early testing, achieving viral suppression and key populations such as men who have sex with men and people who use drugs.[26 27] Additionally, Canada and the USA rely primarily on census-based data collection systems through periodic nationwide statistical surveys. In contrast, some European countries like those in Scandinavia or the Netherlands use national register-based systems which allow for ongoing data collection through existing administrative sources.[28] It is important to note key differences between the Canadian and US health systems: Canada has universal healthcare, whereas the USA does not. These key differences may influence trends in health outcomes due to differences in access and quality of care.[29]

Search strategy and title/abstract screening

To identify potentially relevant papers, we searched two widely used electronic databases, PubMed and Web of Science, combining the term 'HIV' and a comprehensive list of area-based measures of SES (online supplemental table 1). The initial search was conducted from October 2023 to February 2024 and limited to studies published between 1 January 2012 and 31 December 2023. In February 2026, the search period was extended to include studies published between 1 January 2024 and 31 December 2025.

Search results were filtered using language and year of publication in both databases. Additionally, Web of Science allowed for filtering by country, while PubMed search results were excluded manually by reviewers based on country of publication. All search results were exported to Covidence, a software designed to efficiently conduct systematic reviews, for screening and data extraction.[30] Covidence automatically removed duplicates across both databases.

To ensure consistency, all uploaded papers underwent title and abstract screening by two out of three reviewers (NG, NN, KWK). An additional reviewer joined for the extension of the search period (SH). All relevant studies were then moved to the full-text review folder on Covidence. A paper advanced to the data extraction stage if two reviewers agreed on its adherence to the inclusion criteria; the reason for excluding a paper also had to be agreed on. Conflicts were resolved in weekly meetings through team discussions with the third team member, who made the final decision whether to include or exclude the paper in conflict. The data extraction form was jointly developed on Covidence and continuously modified in an iterative process to ensure all relevant variables were extracted. The data from each paper were charted by a single reviewer and all concerns regarding data extraction were discussed at team meetings among all three reviewers. We extracted data on study characteristics (eg, author, year of publication, study design, source of data, type of data used, study sample size), area-based measures (eg, area-based SES measure used and SES dimensions captured by indices), health outcomes assessed and association between area-based SES and health outcomes.

Synthesis

For all included papers, we summarised the study design, the type of area-based SES measures used, the health outcomes assessed and the overall findings on the association between area-based SES and health outcomes. We also examined the different SES domains assessed across studies, providing a count of each domain identified.

Patient and public involvement

As the study is a scoping review, there are no participants. There was no patient or public involvement in the design or conduct of this review.

RESULTS

Study selection

Figure 1 depicts the flow diagram for study inclusion and exclusion. We identified 3470 potentially relevant citations from PubMed and Web of Science using predefined search terms, which were uploaded to Covidence for screening.[30] Duplicate citations were removed either automatically by Covidence (n=1077) or manually (n=2), leaving 2391 citations for title and abstract screening. Of these, 2263 were excluded. Full-text review was performed for the remaining 128 studies, of which 73 were excluded. The most common reasons for exclusion were irrelevant outcomes (n=31) and irrelevant patient populations (n=17) (figure 1). One additional paper was identified through an ancestral citation search and met all inclusion criteria. In total, 56 studies were included in the scoping review.

Study design

Key information extracted from each study is summarised in online supplemental table 2. Of the 56 studies reviewed, most were conducted in the USA (n=53; 95%) and most studies focused solely on PLWH (n=46; 82%). Study designs included retrospective administrative data studies/virtual cohorts (n=39; 70%), such as the Canadian Observational Cohort Collaboration (CANOC) and the Florida electronic Health Assessment Resource System (eHARS) as well as prospective observational cohorts (n=17; 30%), such as the Women's Interagency

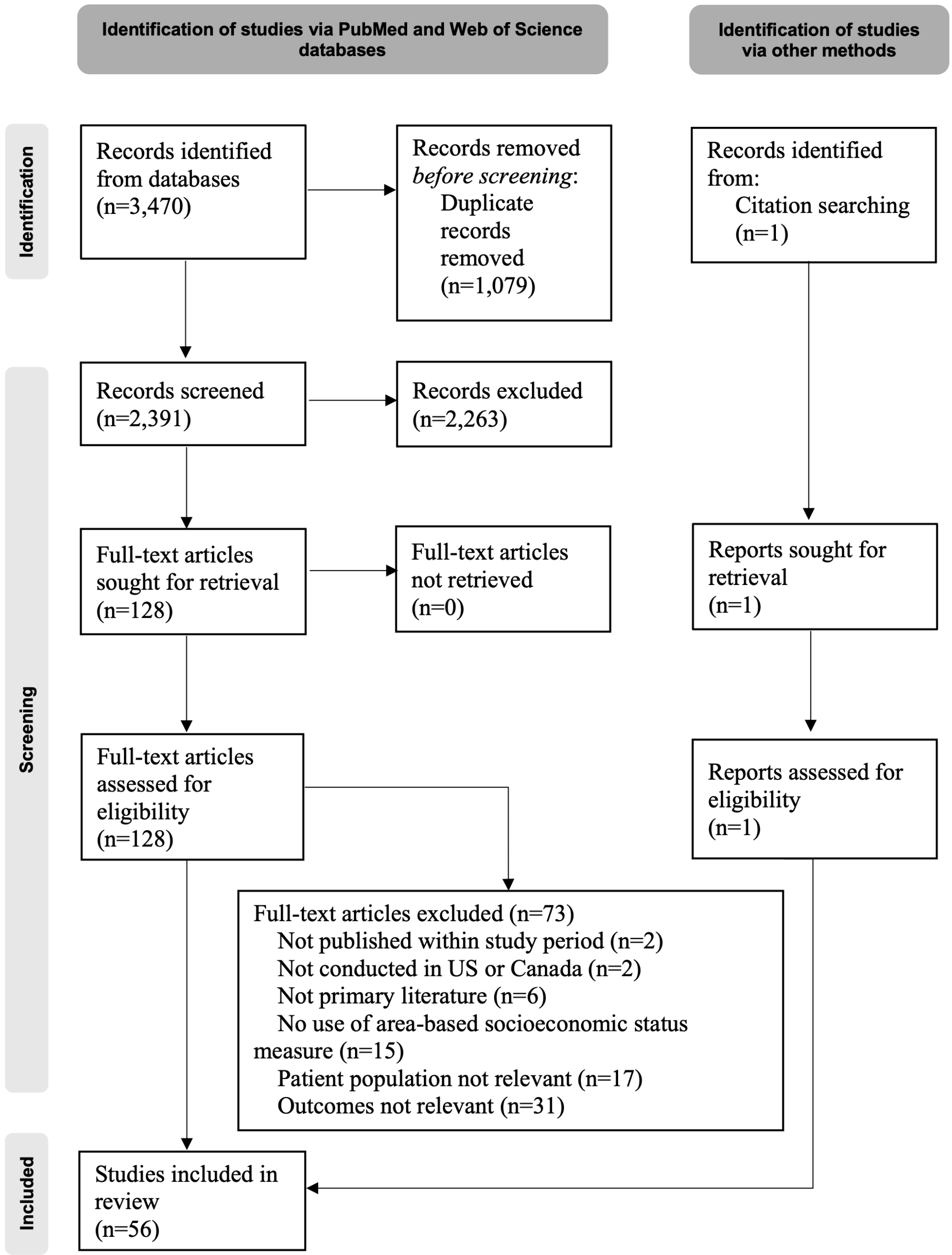

Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) flow diagram for study inclusion and exclusion.

HIV Study (WIHS) cohort.[31–33] Although some prospective observational cohorts were linked to administrative data, we still classified them as such. Several papers (n=22; 39%) conducted cross-sectional studies within either type of these cohorts.

Area-based measures of SES

All 56 studies derived area-based SES measures using national census data, through the US Census Bureau or the Census of Population in Canada.[34 35] Area-based SES was assessed using single indicators directly from census data (n=16; 28%), composite measures (n=34; 61%) or both (n=6; 11%). Composite measures varied based on the combination of indicators used, the weighting assigned to each indicator and geographic area unit in which the measure was calculated. A subset of studies used validated composite measures of SES such as the MSDI (n=2; 4%) and the ADI (n=7; 12%).[21 22] Some composite measures were commonly used with a specific area-unit. For example, ADI was primarily linked with census tract area; however, it can also be linked to other area-units.

The most common geographic area-units were census tracts, which were used in 19 studies (34%). In the USA, census tracts are one of the smallest units, typically including 2500 to 8000 people, and may most closely represent individual SES.[36] In Canada, Dissemination Areas (DA) are the smallest distinct geographic units available, containing 400–700 households, for the purpose of demographic analysis.[37] This was used in the three Canadian studies included (5%).

Chicago community areas, used in one study (2%), are aggregations of census tracts specific to the city of Chicago, with the population ranging from approximately 2200–102 000 people.[38] Similarly, the New York United Hospital Fund areas, also used in one study (2%), are specific to New York City and group the city into 42 neighbourhoods for data presentation.[39] These areas are based on contiguous ZIP codes and in 2010, the median population was 128 117 (Q1, Q3: 83 451, 162 871).[34]

In contrast to the approximately 85 000 census tracts defined for the 2020 census, there are around 41 000 ZIP codes in the USA.[40 41] ZIP code areas are based on postal delivery routes and are not designed for geographic analysis; however, 10 studies (18%) used them to approximate location.[36] Postal codes are the Canadian equivalent of ZIP codes and in 2010 there were approximately 840 000 unique postal codes in Canada.[42] The second most common area-unit used were ZIP code tabulation areas (ZCTAs), which were used in 16 studies (28%). ZCTAs are constructed by the US Census Bureau for statistical analysis and are aggregations of census blocks that share a ZIP code.[43] Census blocks, used in two studies (3%), are the smallest geographic unit used by the US Census Bureau, which divide census tracts into even smaller areas.[44] Some census blocks may have very few residents, resulting in issues with privacy which may limit their use.

Counties, used in three studies (5%), are large area-units that cover areas in the USA with widely varying population depending on rurality and density. In 2024, counties ranged from 48 residents in Loving County, Texas, to 9.7 million in Los Angeles County.[45] Core-based statistical areas (CBSA), used in one study (2%), include both metropolitan and micropolitan areas.[46] A CBSA consists of at least one county with a core urban area with 10 000 or more residents and may include surrounding counties that are strongly integrated. CBSA populations can range from 10 000 to over 22 million, as with the New York-Newark-Jersey City metropolitan statistical area. One study used the city-area level; however, it is unclear how exactly this was defined.[47]

Domains of SES

We adapted frameworks established by the WHO and Healthy People 2030 to categorise the observed types of indicators in the studies included into eight domains: income and poverty (n=56; 100%), education (n=41; 73%), employment and/or occupation (n=38; 68%), social and demographic structure (n=27; 48%), income disparity (n=17; 30%), housing (n=15; 27%), health and well-being (n=9; 16%) and physical living environment (n=6; 11%).[48 49] A description of the eight domains and examples of indicators included in each are provided in table 1. The distribution of area-based SES domains assessed across the reviewed studies is visualised in figure 2.

Health outcomes

Studies assessed measures related to the HIV care cascade or HIV disease progression as health outcomes including viral load/suppression (n=29; 52%), care linkage/retention (n=9; 16%), CD4 count (n=6; 11%) and ART adherence (n=4; 7%). Studies focusing on other health outcomes included mortality (n=8; 14%), sexually transmitted infections (n=4; 7%), mental/emotional distress (n=1; 2%), receptive language (n=1; 2%), alcohol use (n=2; 4%), inflammation and markers (n=1; 2%), body composition (n=1; 2%), brain-age-gap (n=1; 2%), COVID-19 related hospitalisation (n=1; 2%), receiving cancer treatment (n=1; 2%), suicide ideation (n=1; 2%), frailty (n=1; 2%) and liver related events (n=1; 2%).

Most studies (n=46; 82%) found a link between lower area-based SES and poorer health outcomes in PLWH either through direct associations or via attenuation of effects after adjusting for area-based SES. However, 10 studies (18%) did not observe this relationship and highlighted factors which may explain these findings. One study reported limited variability in area-based SES at the census tract level within the sample may explain the lack of association observed.[50]

In some populations, strong community ties or cultural factors may mask some of the negative effects of low SES. For instance, Haitian-Americans with low SES in Florida had better HIV diagnosis outcomes due to strong social networks compared with other populations in Florida,[51] and among Hispanic-Americans living with HIV in Florida, low SES was not linked to retention in care and mortality, likely due to community cohesion.[52 53] In another study, country of birth was found to be associated with linkage to care or retention in care: those born in Mexico or South America had higher prevalence of non-linkage to care and those born in Mexico, South America and the Caribbean had lower prevalence of non-viral suppression.[54]

Additionally, programmes that target resources to areas of higher deprivation, such as in New York City where such a programme was implemented, may weaken some of the link between area-based SES and viral suppression.[55] Moreover, in one study, rural residence was found to be protective against poor retention in care and viral suppression among people living in areas with high deprivation scores, highlighting the effect of rurality.[56]

Some studies found indirect impacts of SES. For example, SES was found to influence ART adherence through self-efficacy,[57] and neighbourhood poverty affected language skills only among youth whose caregivers perceived the neighbourhood as violent or stressful.[58] Additionally, one study found that neighbourhood poverty was not linked to mortality among Hispanic-Americans with a history of injection drug use but was a risk factor for those without.[53] One study found that although SES was not associated with overall frailty status, it was associated with one of the frailty components: those living in more deprived areas reported lower physical activity due to health limitations.[59] These findings further highlight the importance

Table 1 Area-based SES domains assessed across studies: definitions and examples of indicators as identified in the 56 studies included in this review

Domain	Definition	Examples
Income/poverty (n=56)*	Measures the proportion of the population in poverty in an area or indicators of financial resources and accumulated wealth	% poverty, median family income and vehicle ownership
Education (n=41)*	Measures formal education levels and access to educational opportunities, such as the highest level of education achieved, literacy rates and the percentage with a high school diploma	% with less than a high school degree, % of the population aged 25 years or older with less than 9 years of education
Employment and/or occupation (n=38)*	Measures (un)employment rates, job stability and/or the nature of occupations	% of population unemployed, % employed in white collar occupations (16 years and older)
Social and demographic structure (n=27)*	Captures factors related to social organisation and community cohesion	% of households with more than one person per room, % of single parent households with kids under 18 years old and % of population that is female
Income disparity (n=17)*	Direct measures of income inequality in a defined area	High-income to low-income population ratios, and the Gini index, which has been previously used as a factor to predict health outcomes such as HIV infection.[71]
Housing (n=15)*	Captures factors related to housing conditions	Median home value, median gross rent, median monthly mortgage, % of owner-occupied housing units, home ownership vs renting
Health and well-being (n=9)*	Measures physical and mental health outcomes, as well as access to healthcare services	Gallup Well-Being Index, which has previously been used to explore health outcomes and insurance coverage.[72] Prevalence of sexually transmitted infections, prevalence of other diseases and insurance coverage
Physical living environment (n=6)*	Captures aspects of the physical environment which can impact quality of life	Violent crime rate, neighbourhood density, distribution of alcohol outlets in an area

*Number of studies that included this domain in their area-based SES measure.
SES, socioeconomic status.

of examining mediating and moderating factors to better understand the impact of SES on health outcomes among PLWH.

DISCUSSION

Area-based measures of SES are being increasingly used in HIV research, reflecting a broader shift in public health toward understanding health as socially and structurally produced. As census infrastructure and administrative health data linkages have expanded, researchers may have wider access to geographically derived indicators to approximate the socio-structural conditions shaping health among PLWH. This scoping review synthesised 56 studies published between 2012 and 2025 in Canada and the USA assessing health outcomes among PLWH, providing a comprehensive overview of area-based measures of SES used, their methodological application and their implications for health equity. Our findings indicate that area-based SES measures, exclusively derived from census data, were consistently associated with poorer health outcomes, reinforcing the socio-structural influence of health inequities among PLWH. There was considerable heterogeneity in how SES was operationalised, including the domains that were selected and the area-units that were applied.

Domains of SES

Our review found that studies most often included indicators of income/poverty, education and/or employment. Social domains such as housing, health and well-being and factors related to the physical living environment were comparatively under-measured. This imbalance suggests that the conceptualisation of socio-structural disadvantage in HIV research is dominated by measures of material rather than social deprivation.

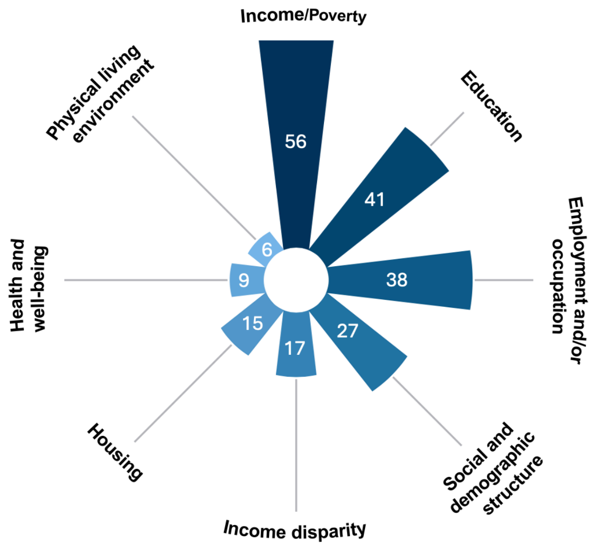

Figure 2 Number of studies using each area-based socioeconomic status (SES) domain among the 56 included studies.

A subset of studies used validated composite indices such as the ADI (comprising 17 different indicators of material and social condition) and MSDI (captures multiple factors such as adequate housing, possession of a car, living in a neighbourhood with recreational areas, individuals living alone, being a lone parent and being separated, divorced or widowed).[21 60] The use of validated composite indices enhances consistency and reproducibility across studies. However, these indices were not developed specifically for HIV research; therefore, they may lack the flexibility to account for relevant local context or population-specific factors, and their composition may change over time as census variables evolve, which is important to consider for longitudinal studies.

Modifiable areal unit problem and conceptualising area-based SES

A central methodological issue identified in our findings is the modifiable areal unit problem (MAUP), which describes the bias introduced by the choice of geographic boundaries which may influence observed associations.[61] Area-based SES measures capture contextual exposures reflecting socio-structural conditions, rather than individual socioeconomic circumstances. Therefore, the choice of area-unit should be guided by the hypothesised mechanisms linking the setting and health in each research question. Researchers should clearly justify their choice of area-units used and communicate how this may inform the interpretation of their findings. In studies considering broader policy-level determinants of healthcare access, such as health insurance coverage, larger administrative regions (eg, CBSA, county) aligned with healthcare delivery systems may be more relevant.[62] Smaller area-units (eg, census tract, ZCTA and DA) capture the most granular area conditions and may better approximate the socioeconomic environments an individual experiences.[63] These considerations can help reduce misinterpretation and reduce potential bias related to MAUP.

Framework for constructing and combining area-level and individual-level measures of SES

Area-based measures, such as median household income in an area, and individual-level measures of SES, such as self-reported income, reflect related but distinct dimensions of social and material context.[16] Researchers aiming to understand the independent, additive and/or interactive effects of both individual factors and environmental context should explore using both levels of SES measures. In addition to applying a theory-driven selection of SES domains at both individual and area levels, studies should consider complementary analytic approaches that include an assessment of the agreement between individual and area-based measures of SES through descriptive comparisons, and/or parallel models.[16]

The use of multiple models may be a particularly useful methodological approach for combining individual and area-based SES measures. For example, analyses may begin with separate models adjusting for individual-level and area-level SES, followed by combined models including both measures and their interaction.[18] This approach can help determine whether associations persist, attenuate or interact when both levels are considered; can help clarify potential mechanisms and reduce the risk of obscuring SES effects; and allows for the examination of cross-level interactions, such as whether the impact of low individual income varies according to neighbourhood deprivation.

Area-level SES measures as a proxy for individual level markers

In some settings, particularly large population-based cohorts where individual level SES may not be widely available, area-level measures of SES can serve as a reasonable proxy if carefully selected and interpreted. For example, a recent study from the North American AIDS Cohort Collaboration on Research and Design found ZIP code-based SES indicators (income, education, and employment) were associated with viral suppression among PLWH in North America, producing patterns comparable to studies using individual-level SES.[64] This study by Chandran *et al* outlines the importance of a thoughtful and conceptually grounded approach to selecting area-based measures SES indicators as surrogates for individual-level SES.[64] Specifically, the selection of SES indicators should be guided by an evidence-derived conceptual framework. Researchers should choose an area-unit which produces outcome patterns consistent with studies using individual-level SES in the population being studied. If prior research does not identify a single preferred indicator, multiple indicators such as area-based income, education and employment should be assessed independently and/or in combination. The latter step helps address an inherent limitation of composite SES measures: they may obscure the mechanisms through which structural inequities operate,[65] as illustrated in one study included in this review reporting that area-level poverty explained racial disparities in survival just as well as a composite index.[66]

Broader social and environmental determinants of health among PLWH

Our findings emphasise the importance of considering broader social and environmental contexts, some of which may not be captured by the various area-based SES measures. Given stigma, discrimination and social isolation have been demonstrated to shape the lived experiences of PLWH, factors such as community cohesion, social capital and cultural resilience may play important roles in mediating or modifying associations between SES and health outcomes.[67] In some settings, these can act as compensatory factors, helping PLWH navigate or partially offset some of the negative effects of socio-structural deprivation they may otherwise face. For example, strong social networks and culturally tailored community-based organisations may facilitate access to HIV testing and alternative care services—contributing to improved HIV care outcomes among Haitian-Americans in Florida, USA.[51] Among other Latin-American populations residing in Florida, dense community networks and culturally congruent healthcare providers, such as Spanish speaking clinicians, may contribute to higher retention in HIV care.[52] A study from New York, USA, demonstrated that the presence of accessible and effective healthcare services in low-SES areas may mitigate some of the health outcomes associated with economic deprivation.[54] While these factors do not eliminate socio-structural disadvantage, they can act as compensatory factors, helping PLWH navigate or partially offset some of the negative effects of socio-structural deprivation they may otherwise face. Future research should systematically examine these factors further.

Healthcare system context is relevant when interpreting findings from Canada and the USA. Canada has a publicly funded healthcare system in which access to medical services is considered a universal right. Regional governments run public drug programmes which generally provide universal coverage for combination antiretroviral therapy (cART) and reduce financial barriers to HIV treatment. However, even within a universal healthcare system, differences in healthcare infrastructure, service capacity and quality may contribute to socio-structural barriers in accessing healthcare.[61] In contrast, healthcare in the USA is generally tied to insurance coverage and financial resources, likely amplifying the relationship between SES and health outcomes among PLWH. Programmes such as Medicaid and the Ryan White HIV/AIDS Program may provide financial support for HIV care to some individuals with low income or insufficient insurance coverage.[68] In both countries, rurality may further modify how SES influences health outcomes among PLWH. While HIV services are often more concentrated in urban areas and have been associated with improved treatment outcomes in some settings, such as British Columbia, Canada, the relationship between rurality and SES is context dependent.[69 70] In some regions, rural areas may be more socioeconomically deprived, whereas in others, the most deprived populations are concentrated in urban settings such as inner-city neighbourhoods or areas with high housing instability. Therefore, the effects of rurality and SES should be interpreted within their specific geographic and social context.

Strengths and limitations

This review focused exclusively on studies from Canada and the USA and therefore our results may not be generalisable to other settings. Only three Canadian studies met the inclusion criteria, likely reflecting fewer studies and less available data relative to the USA. Moreover, due to the wide variability in SES measures and terminology, some relevant studies may have been missed. Finally, our review found most studies focused on virologic outcomes, with comparatively less attention to ageing-related comorbidities and mental health conditions; however, in more recent years, there has been a shift towards these outcomes as PLWH increasingly live longer and age, highlighting an important area for future research.

Implications and recommendations for future research

Given our findings, we put forth several recommendations for future research that use area-based measures of SES to assess health outcomes among PLWH. (1) The choice of SES domains should be conceptually anchored and the measurement complexity should be purposeful, (2) the choice of area-units used should be explicitly justified and grounded in an evidence-based theoretical framework, (3) analyses should explore multiple area-based SES measures and/or conduct sensitivity analyses when feasible, (4) researchers should consider combining area-based and individual-level SES measures and (5) researchers should be aware of other factors (such as social capital, cultural resiliency, targeted programme(s), rural/urban dynamics) that influence the relationship of the area-based SES measurement to health outcomes.

CONCLUSION

This scoping review synthesises the use of area-based SES measures used across 56 studies which assess health outcomes among PLWH in Canada and the USA. We found that lower area-based SES is generally linked to poorer health outcomes among PLWH. By mapping the landscape of area-based SES use, this review not only advances methodological approaches in HIV research, but also contributes to the conversation on health equity, laying the foundation for future research and interventions. To strengthen research in this area, we recommend grounding the choice of area-units and SES domains in theoretical frameworks, using composite or simple SES measures purposefully, combining area-level and individual-level measures of SES when feasible, and exploring broader social determinants of health not traditionally captured in area-based measures of SES.

Contributors NN, NG, and KWK designed the study, including the development of the scoping review aim, objective, and protocol with input from all co-authors. NN, NG, and KWK developed the methodology and led the screening, data extraction, and interpretation. SH contributed to screening and data extraction during the extended literature search. NN was responsible for data visualisation. NN, NG, and KWK drafted the initial manuscript. All authors contributed to interpreting the findings, critically revised the manuscript, and approved the final version. RSH acted as the guarantor.

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